Computational Statistics and Simulation: Section 2. Monte Carlo methods in Statistical Inference

Master in Mathematics and Applications

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Section 2: Monte Carlo methods in Statistical Inference

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1. Introduction

Generically, Monte Carlo methods are all computational tools where one examines properties of a probability distribution, generating a large sample of a given distribution and then the sample statistical properties are studied.

In this chapter we will consider several Monte Carlo Methods in order to numerically approximate the average value $E[g(X)]$ for a real g function.

Definition of mean value $E[g(X)]$

Let X be a continuous rv with pdf f and g a real function. The mean value of $g(X)$, $E[g(X)]$, is defined by

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f(x) dx \quad (1)$$

2. Monte Carlo integration: definition

If the integral in (1) cannot be solved analytically, one can try to get an approximation to the value of the integral.
Different methods to get approximations for $E[g(X)]$:

- using numerical integration to get an approximation to the value of the integral.
  Numerical integration is outside the topic of statistics computation, so we will not address this approach.
- through Monte Carlo integration technique

In this course we will study Monte Carlo integration; which is based on the

(Strong) Law of Large Numbers: $E[g(X)] = lim \frac{1}{N} \sum_{i=1}^{N} g(X_i)$ (2)

Although result (2) is valid when $N \to \infty$, in practice, we use

$$E[g(X)] \approx \frac{1}{N} \sum g(X_i)$$

for large values of N, with X_i being generated by the pdf f.

2. Monte Carlo integration: algorithm

The term "Monte Carlo"(MC) is a reference to the casino of Monte Carlo (Roulette tables are random number generators in a sense), and the word "integration"refers to the integral (1).

Monte Carlo integration algorithm

input: g function, $n \in N$

PRN's: iid observations $X_1, \ldots, X_N$ with distribution X

output: estimator for $E[g(X)]$

1. $s := 0$
2. for $i = 1, 2, \ldots, N$ do
3. $s := s + g(X_i)$
4. **end for**
5. return $s = (s + g(X_i))/N$

2. Monte Carlo integration: algorithm

Summaryzing:

Monte Carlo Method (2)

Suppose we you want to calculate $\theta = \int g(x)f(x)dx$

So $\theta = E[g(X)]$ where X has probability density function f(x).

If we have $X_1,\ldots,X_n$ with density f(x), then θ can be estimated using the Monte Carlo estimator

$$\hat{\theta} \approx \frac{1}{n}\sum_{i=1}^{n} g(x_i)$$

Algorithm:

1. Generate $X_1,\ldots,X_n$ with density f(x)
2. Compute $\hat{\theta} = \frac{1}{n}\sum_{i=1}^{n} g(X_i)$
3. Set $\hat{\theta}$ for an approximation of θ

Example 1: Monte Carlo integration

Calculate Monte Carlo estimate for

$$\theta = \int_0^1 e^{-x} dx$$

and compare with the exact value

In R the algorithm can be written as

```
> m <- 10000
> x <- runif(m)
> theta.hat <- mean(exp(-x))
> print(theta.hat)
[1] 0.6321036
> print(1 - exp(-1))
[1] 0.6321206
```

2. Monte Carlo integration: examples

Using MC integration how can we approximate integrals defined in a and b, with the form

$$\int_a^b g(x)dx?$$

Let U_i iid with $U \sim U(a,b)$, i.e, $f_U(x) = \frac{1}{b-a}\mathbf{1}_{(a,b)}(x)$

Then, for large values of N

$$\int_a^b g(x)dx = (b-a)\int_a^b g(x)f_u(x)dx = (b-a)E[g(U)] \approx \frac{b-a}{N}\sum_{i=1}^{N} g(U_i)$$

Example 2: Monte Carlo integration

Calculate Monte Carlo estimate for

$$\theta = \int_2^4 e^{-x} dx$$

and compare with the exact value

In R the algorithm can be written as

```
>  m <- 10000
>     x <- runif(m, min=2, max=4)
>     theta.hat <- mean(exp(-x)) * 2
>     print(theta.hat)
[1] 0.1176214
>     print(exp(-2) - exp(-4))
[1] 0.1170196
```

2. Monte Carlo integration: examples

Using MC integration how can we approximate probabilities?

Rewritten them as mean values

If X is a r.v. and X_i are iid with X, then for large value of N

$$P(X \in A) = E(\mathbf{1}_A(X)) \approx \frac{1}{N} \sum \mathbf{1}_A(X_i)$$

Example 3: Monte Carlo integration
Let $X \sim N(0,1)$.
Calculate a Monte Carlo estimate for

$$P(X \leqslant 4)$$

Although it is not possible to get it analytically, a MC estimator is

$$P(X \leqslant 4) \approx \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}_{\{X_i \leqslant 4\}}$$

with $X_1, \ldots, X_N$ generated from a normal distribution $N(0,1)$, and N large.

Example 4: Monte Carlo integration

Let $X \sim N(0,1)$.

Calculate a Monte Carlo estimate for

$$P(X \geqslant 4.5)$$

Approach 1

Following the same procedure

$$P(X \geqslant 4.5) \approx \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}_{\{X_i \geqslant 4.5\}}$$

with $X_1, \ldots, X_N$ generated from a normal distribution $N(0,1)$, and N large.

However, this method requires a very large number of PRN's N(0,1), since it is very close to 0.

2. Monte Carlo integration: examples

Indeed, only for values of the order of $N = 10^7$, it is possible to obtain estimates with some precision; running the algorithm in R:

```
a=4.5
 for(i in 5:7){
     N=10^i
     z<-rnorm(N)
     p.hat=sum(z>4.5)/N
     p= pnorm(-a)
     dif=abs( sum(z>4.5)/N- pnorm(-a))
     mylist<-list(N=N,p.hat=p.hat,p=p,dif=dif)
     print(mylist)
}
}
```

uma saída possível é a seguinte:

```
$N
[1] 1e+05

$p.hat
[1] 0

$p
[1] 3.397673e-06

$dif
[1] 3.397673e-06

$N
[1] 1e+06

$p.hat
[1] 2e-06

$p
[1] 3.397673e-06

$dif
[1] 1.397673e-06

$N
[1] 1e+07

$p.hat
[1] 3.3e-06

$p
[1] 3.397673e-06

$dif
[1] 9.767312e-08
```

2. Monte Carlo integration: examples

For $X \sim N(0,1)$ the events of the form $X > 4.5$ can be "cataloged" of **rare events**; we can overcome this situation considering an **auxiliary** model with pdf g(x) of easy generation, and with localization in this support.

Approach 2: considering auxiliary models

Possibility 1:
auxiliary model: **unit exponential shifted to 4.5**

$$g(x) = e^{-(x-4.5)}, x > 4.5$$

Then, the MC estimate can be given by

$$P(X > 4.5) \approx \frac{1}{N} \sum_{i=1}^{N} \frac{\phi(x_i)}{g(x_i)}$$

with $\phi \hookrightarrow$ pdf of $N(0,1)$
$g \hookrightarrow$ pdf of unit exponential shifted to 4.5, with distribution function $G(x) = 1 - e^{-(x-4.5)}$

2. Monte Carlo integration: examples

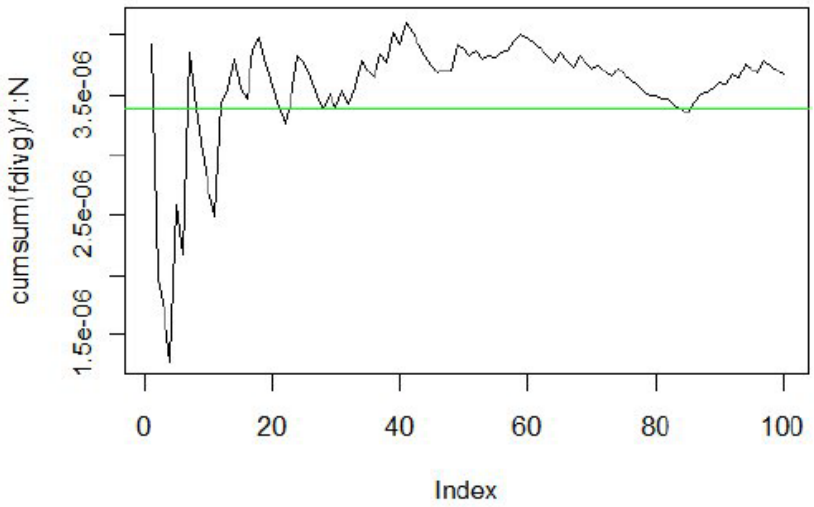
Implementing the algorithm in R:

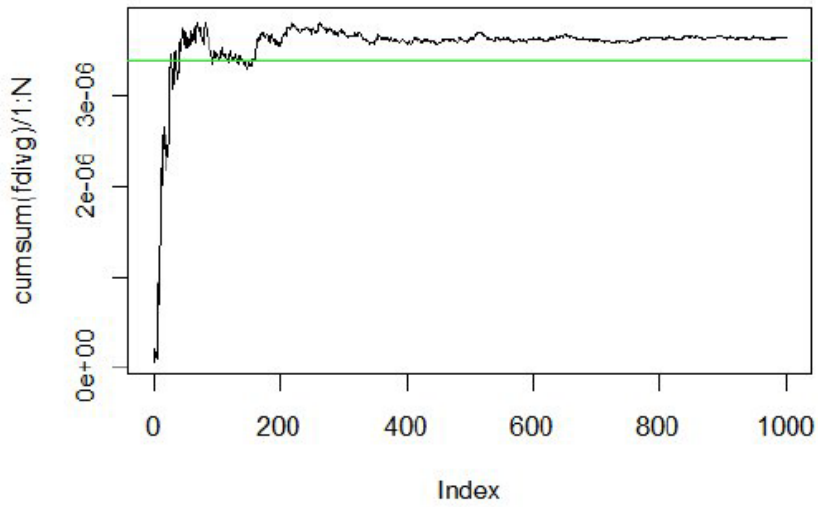
```
par(mfrow = c(2, 2))
a=4.5
for(i in 2:5){
N=10^i
y=rexp(N)+a
fdivg=dnorm(y)/dexp(y-a)
vect<-cumsum(fdivg)/1:N
plot(cumsum(fdivg)/1:N,type="l")
abline(h=pnorm(-a),col="green")  # probabilidade P[Z>4.5]=P[Z<-4.5]
cat("N=",N,"p.hat=",vect[N]," p=",  pnorm(-a),
  "dif=", abs(vect[N]-pnorm(-a)),"\n")
}
}
```

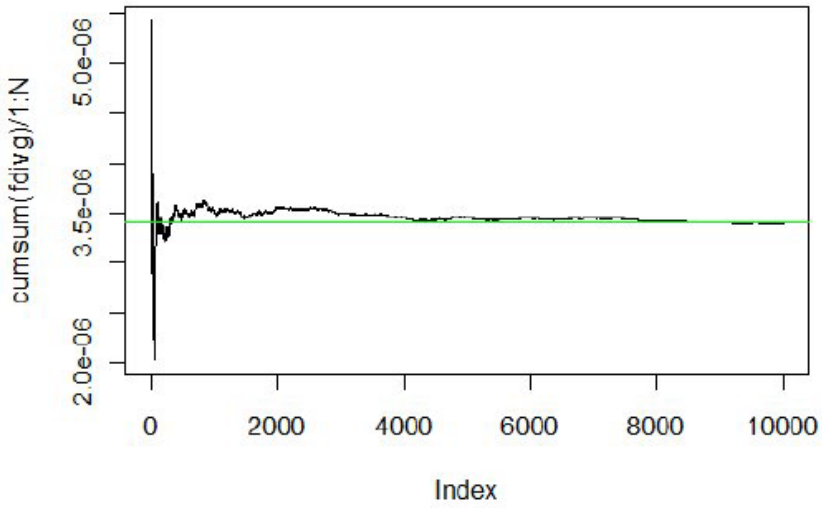
possible output:

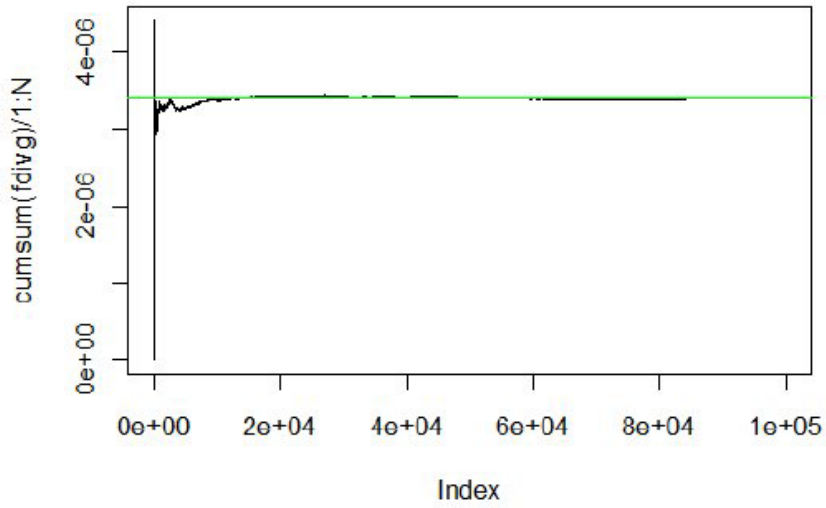
```
at= 3.678575e-06 p= 3.397673e-06   dif= 2.809018e-07
N= 1000 p.hat= 3.640066e-06 p= 3.397673e-06   dif= 2.423931e-07
N= 10000 p.hat= 3.390207e-06 p= 3.397673e-06   dif= 7.466403e-09
N= 1e+05 p.hat= 3.398206e-06 p= 3.397673e-06   dif= 5.328124e-10
```

2. Monte Carlo integration: examples









Approach 2: considering auxiliary models

Possibility 2:
auxiliary model: **shifted Pareto to 4.5**

$$g(x) = \alpha(x - 3.5)^{(} - 1 - \alpha), x > 4.5$$

Then, the MC estimate can be given by

$$P(X > 4.5) \approx \frac{1}{N} \sum_{i=1}^{N} \frac{\phi(x_i)}{g(x_i)}$$

with $\phi \hookrightarrow$ pdf of $N(0, 1)$
$g \hookrightarrow$ pdf of shifted Pareto to 4.5, with distribution function
$G(x) = 1 - (x - 3.5)^{(-\alpha)}$

Exercise: considere $\alpha > 5$ and compare with the shifted exponential as auxiliary model

Open question: how large should N be chosen?

- On the one hand, the larger N is, the more accurate will be the MC estimate; since $N \to +\infty$ MC approximation will converge to the right value.

- But on the other hand the larger N, the more expensive the method, because we need to generate and process more PRN's values X_i.

A key to choice of N is based on the following two results:

2. Monte Carlo integration: estimation accuracy

- MC estimator is unbiased for $E[g(X)]$, i.e.,

$$E[\frac{1}{N}\sum_{i=1}^{N} g(X_i)] = \frac{1}{N}E[\sum_{i=1}^{N} g(X_i)] = E[g(X)]$$

- The variance of the MC estimator is

$$V[\frac{1}{N}\sum_{i=1}^{N} g(X_i)] = \frac{1}{N^2}V[\sum_{i=1}^{N} g(X_i)] = \frac{V[g(X)]}{N}$$

ie, the variance goes to 0 when $N \to \infty$, (consistency), decaying to zero at speed $1/\sqrt{N}$; that is, to increase the precision by a factor of 10, ie, by one more significant digit, is required to increase the # of X_i's generated by a factor of 100.

2. Monte Carlo integration: estimation accuracy

To determine the necessary value N necessary to achieve a given precision, one can proceed as follows: by the Chebyshev inequality,

$$P\left[\left|\frac{1}{N}\sum_{i=1}^{N} g(X_i) - E[g(X)]\right| \leq \varepsilon\right] \geq 1 - \frac{Var(g(X))}{N\varepsilon^2}.$$

Taking $\alpha = \frac{Var(g(X))}{N\epsilon^2}$, choosing $N \geqslant \frac{Var(g(X))}{\alpha\epsilon^2}$, we get

$$P\left[\left|\frac{1}{N}\sum_{i=1}^{N} g(X_i) - E[g(X)]\right| \leq \varepsilon\right] \geq 1 - \alpha.$$

Suppose we have a majorant for the variance $Var(g(X)) \leqslant 1$, for example; for example; to estimate $E[g(X)]$ with maximum error $\epsilon = 0.01$ with a probability of at least $1 - \alpha = 0.95$, will have to use

$$N \geqslant \frac{1}{\alpha\epsilon^2} = \frac{1}{0.05(0.01)^2} = 200000$$

simulated values.

2. Monte Carlo integration: general situation

Generically, to get a MC estimate of the integral

$$I = \int_{\chi} h(x)dx,$$

I must express a mean value; so, we can make the decomposition

$$h(x) = g(x)f(x)$$

with $f(x)$ being the pdf of a rv with support χ, then

$$I = \int_{\chi} h(x)dx = \int_{\chi} g(x)f(x)dx = E[g(X)].$$

2. Monte Carlo integration: general situation

To get a MC estimate of the integral I, we simulate N PRN's observations from the model X: $x_1, x_2, \ldots, x_N$;

the simulated value that gives an approximation to I is given by

$$I \approx \hat{I} = \frac{1}{N} \sum_{i=1}^{N} g(x_i).$$

We have seen

$$\sqrt{\frac{V(g(X)}{N}} \hookrightarrow \text{Monte Carlo estimation error}$$

that will be estimated through its empirical counterpart:

$$\sqrt{\frac{s^2_{g(x)}}{N}} \hookrightarrow \text{estimate of the Monte Carlo error}$$

where

$$s^2_{g(x)} \hookrightarrow \text{denotes the sample variance of the simulated values}$$

$$g(x_1), \ldots, g(x_N)$$

2. Monte Carlo integration: importance sampling

Importance sampling method approximates the integral

$$I = E[g(X)] = \int_{\chi} g(x)f(x)dx$$

carrying out an alternative representation:

$$I = E[g(X)] = \int_{\chi} g(x)f(x)dx = \int_{\chi} \frac{g(x)f(x)}{h(x)}h(x)dx.$$

Then, we obtain the estimate given by Monte Carlo simulation

$$\hat{I} = \frac{1}{N}\sum_{i=1}^{N} \frac{g(x_i)f(x_i)}{h(x_i)}$$

with x_i being generated from pdf $h(x)$

An advisible choice that h model should obey is:

- the support of h should contain the support of f
- h tails should be more heavy than f tails

3. Applications in Statistical Inference

- In statistical inference usually we have a sample

$$X_1, X_2, \ldots, X_n$$

that we assume being generated according to some statistical model.

Goal: to get information about the parameters of the model.

- Let

$$T = T(X_1, \ldots, X_n) \hookrightarrow \textit{statistics}$$

$T \hookrightarrow$ estimator, if it is used to estimate a parameter.

- Often, especially for small n, the distribution of T is not explicitly known, but we can study it using the Monte Carlo method.

We have an unbiased estimator T of parameter μ, i.e, $\mu = E(T)$

i)How can we estimate μ?

Using replicate method to get a **point estimate:**

- Generate N independent samples of the model (N is the number of replications):

$$(X_1^{(1)}, X_2^{(1)}), \ldots, X_n^{(1)}), \ldots, (X_1^{(N)}, X_2^{(N)}, \ldots, X_n^{(N)})$$

- Calculate for each simulated sample the value of the statistics T:

$$T^{(1)} = T(X_1^{(1)}, \ldots, X_n^{(1)}), \ldots, T^{(N)} = T(X_1^{(N)}, \ldots, X_n^{(N)})$$

- Approximate $\mu = E[T]$, by $\bar{T}$, ie:

$$\mu \approx \hat{\mu} = \bar{T} = \frac{1}{N} \sum_{i=1}^{N} T^{(i)}$$

ii)How can we get an approximation to the confidence interval for $\mu = E[T]$?

From Slutsky theorem, we know that:

$$\sqrt{N}\frac{\bar{T}-\mu}{S_T} \sim N(0,1)$$

where:

- $S_t^2 = \frac{1}{N}\sum_{i=1}^{N}(T^{(i)} - \bar{T})^2 \hookrightarrow$ sample variance of $(T^{(1)}, \ldots, T^{(N)})$
- $z_{1-\frac{\alpha}{2}} \hookrightarrow$ quantil of order $(1-\frac{\alpha}{2})$ of $N(0,1)$ distribution

Then

$$P(-z_{1-\frac{\alpha}{2}} \leqslant \sqrt{N}\frac{\bar{T}-\mu}{S_T} \leqslant z_{1-\frac{\alpha}{2}}) \approx 1-\alpha$$

Leading to the **confidence interval approximation**, with $100(1-\alpha)\%$:

$$(\bar{T} - z_{1-\frac{\alpha}{2}}\frac{S_T}{\sqrt{N}}, \bar{T} + z_{1-\frac{\alpha}{2}}\frac{S_T}{\sqrt{N}})$$

iii) Approximation to the distribution function of T

$F_T(x) = P(T \leqslant x)$, can be estimated by the correspondent empirical distribution function, i.e.,

$$F_T(x) \approx \hat{F}_T(x) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}_{\{T^{(i)} \leqslant x\}}$$

iv) Approximation to the p order quantile of T distribution, $\chi_{p,T}$:

$\chi_{p,T} : P(T \leqslant \chi_{p,T}) = p$

Can be estimated through a long run of N simulated values of T statistics:

$$T^{(1)}, \ldots, T^{(N)}$$

ordering them : $T_{(1)} \leqslant T_{(2)} \ldots \leqslant T_{(N)}$.

The p-order quantile of T distribution is estimated by:

$$\chi_{p,T} \approx \hat{\chi}_{p,T} = T_{([Np]+1)}$$

with $[Np] \hookrightarrow$ integer part of Np.

iv)How to get an approximation to the confidence interval for p-order quantile of T?

An approximate $(1-\alpha)$ confidence interval for the p-order quantile of T is

$$(T_{(l)}, T_{(u)})$$

where:

- $l = [Np - \sqrt{Np(1-p)}z_{1-\alpha/2}]$

  $u = [Np + \sqrt{Np(1-p)}z_{1-\alpha/2}]$

- $z_{1-\alpha/2} \hookrightarrow (1-\alpha/2)$ Quantile of $N(0,1)$

Justification: Considering a continuous T variable and $l < u$

$$1-\alpha \approx P(T_{(l)} \leqslant \chi_{p,T} \leqslant T_{(u)}) = \sum_{i=l}^{u-1} C_i^N p^i (1-p)^{N-i}$$

and $B(N,p) \approx N(Np, Np(1-p))$, for large N.

Suppose that the statistics T is a estimator of parameter θ, i.e.,

$$T = \hat{\theta} = \hat{\theta}(X_1, \ldots, X_n)$$

v)How to get an approximation to the bias of the estimator $\hat{\theta}$?

Def: $Bias(\hat{\theta}) := E[\hat{\theta} - \theta]$

It can be estimated by: $\hat{Bias}(\hat{\theta}) = \frac{1}{N}\sum_{i=1}^{N} \hat{\theta}^{(i)} - \theta$

vi)Approximation to the Mean Square Error (MSE) of $\hat{\theta}$?

Def: $MSE(\hat{\theta}) := E[(\hat{\theta} - \theta)^2]$

It can be estimated by: $\hat{MSE}(\hat{\theta}) = \frac{1}{N}\sum_{i=1}^{N} (\hat{\theta}^{(i)} - \theta)^2$

where $\hat{\theta}^{(i)} = \hat{\theta}^{(i)}(X_1, \ldots, X_n)$ is the estimator $T = \hat{\theta}$ obtained using the i-th sample replication $(X_1^{(i)}, \ldots, X_n^{(i)}), i = 1, \ldots, N$

Application of MC simulation in Statistical Tests

- Let $T = T(X_1, \ldots, X_n)$ be the statistics of the test that allows to decide the H_0 (null hypothesis) rejection or not.
- R is the rejection region of the test, with significance level α, meaning

$$P(RejectH_0|H_0true) = P(T \in R|H_0true) \leqslant \alpha$$

- In many cases this condition can only be obtained asymptotically- *tests with asymptotic level* α- in practice approximations are used, when n is large.

Then, for small values of n, α is unknown and can be estimated by MC simulation.

Application of MC simulation in Statistical Tests

In order to estimate α for small values of n, using Monte Carlo simulation, we proceed as follows:

- Generate N independent samples according to the model under H_0;

$$(X_1^{(i)}, X_2^{(i)}, \ldots, X_n^{(i)}), i = 1, .2, \ldots, N$$

- Calculate the value of the statistics of the test for each replicate

$$T^{(i)} = T(X_1^{(i)}, X_2^{(i)}, \ldots, X_n^{(i)}), i = 1, 2, \ldots, N$$

- Determine the percentage of samples that leads (wrongly) to the rejection of H_0, meaning

$$\alpha \approx \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}_{\{T^{(i)} \in R\}}$$

Application of MC simulation in Statistical Tests

- For this MC simulation type, recall that the estimated parameter α corresponds to a probability

$$\alpha = p := P(Type1error)$$

- the estimator is the proportion

$$\hat{\alpha} = \hat{p} := \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}_{\{T^{(i)} \in R\}}$$

- Remember that a estimate of the standard deviation of that proportion is

$$\hat{sd}(\hat{\alpha}) = \sqrt{\frac{\hat{\alpha}(1-\hat{\alpha})}{N}} \leqslant \frac{0.5}{\sqrt{N}}$$

Def: Asymmetry coefficient)
The asymmetry coefficient, denoted by $\sqrt{\beta_1}$ for a rv X is

$$\sqrt{\beta_1} = \frac{E[(X - \mu_X)^3]}{\sigma_X^3}$$

with $\mu_X = E[X]$, $\sigma_X = \sqrt{V[X]}$

- If $\sqrt{\beta_1} = 0$, the distribution is symmetric
- If $\sqrt{\beta_1} > 0$, the distribution has positive assimetry
- If $\sqrt{\beta_1} < 0$, the distribution has negative assimetry

In practice the asymmetry coefficient is estimated using the empirical asymmetry coefficient

$$\sqrt{b_1} = \frac{\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2}{(\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^3)^{2/3}}$$

Example: Asymmetry test for the normality

Consider the normality test for a population X based on the asymmetry coefficient,

$$H_0 : \sqrt{\beta_1} = 0 \quad vs \quad H_1 : \sqrt{\beta_1} \neq 0$$

and study the α for small values of n, under normality hypothesis and when considering asymptotic critical points.

We know that: If $X \sim Normal$, then $\sqrt{\frac{n}{6}}\sqrt{b_1} \approx N(0,1)$

The normal distribution is simmetric; a normality test based on the asymmetry rejects normality hypothesis for large values of $\sqrt{b_1}$. Then the hypothesis

$$H_0 : X \sim Normal \quad vs \quad H_1 : X \nsim Normal$$

can be reformulated, based on asymmetry coefficient:

$$H_0 : \sqrt{\beta_1} = 0 \quad vs \quad H_1 : \sqrt{\beta_1} \neq 0$$

Example (cont): Asymmetry test for the normality

With significance asymptotic level α, the rejection region is defined by

$$R : |\sqrt{b_1}| > z_{1-\frac{\alpha}{2}}$$

onde $z_{1-\frac{\alpha}{2}} \hookrightarrow$ quantile of order $(1-\frac{\alpha}{2})$ of a N(0,1)

However, the convergence of $\sqrt{b_1}$ for the limiting distribution is slow and the asymptotic distribution is not a good approximation for moderate values of n.

Example (cont): Asymmetry test for the normality

Let's fix $\alpha = 0.05$; we begin by calculating the critical points for the following values for $n = 10; 20; 30; 50; 100; 500$
In R programme:

```
n <- c(10, 20, 30, 50, 100, 500)
cv <- qnorm(.975, 0, sqrt(6/n))
cv <- round(cv,4)
```

Output of the critical points:

```
n      10     20     30     50    100    500
cv 1.5182 1.0735 0.8765 0.6790 0.4801 0.2147
```

Example (cont): Asymmetry test for the normality

Procedure to get an approximation to significance level:

For each value of $n[i], i = 1, 2, \ldots, 6$, make N=1000 replications of the folllowing steps

- generate a sample from a Normal distribution with size $n[i]$
- calculate the correspondent asymmetry coefficient (function sk)
- define test rule as 1 to rejection or 0 not to reject, and register in a vector of size N
- in the end, calculate the rejection proportion of H_0

In R programme:

Function to calculate the asymmetry coefficient:

```
sk <- function(x) {
       # cálculo do coeficiente de assimetria empírico
       xbar <- mean(x)
       m3 <- mean((x - xbar)^3)
       m2 <- mean((x - xbar)^2)
       return( m3 / m2^1.5 )
     }
```

$n \hookrightarrow$ vector of the sample sizes

length$(n) \hookrightarrow$ sizes of different simulations

```
p.reject <- numeric(length(n)) # armazena resultados da simulação
N <- 10000                     # número replicas para cada simulação

for (i in 1:length(n)) {
       sktests <- numeric(N)     # decisões do teste
       for (j in 1:N) {
           x <- rnorm(n[i])
           # decisão do teste -- 1 (rejeição) ou 0
           sktests[j] <- as.integer(abs(sk(x)) >= cv[i] )
           }
       p.reject[i] <- mean(sktests) # proporção de rejeições
   }

> p.reject
[1] 0.0117 0.0255 0.0367 0.0417 0.0451 0.0520
```

Example (cont): Asymmetry test for the normality

- The simulation results for α=P(Type 1 error), using the asymptotic critical points is summarized in the table below:

n	10	20	30	50	100	500
probERR01	0.0117	0.0255	0.0367	0.0417	0.0451	0.0520

- For $N = 100000$ the standard deviation for each estimate of probability α is approx. $\sqrt{\frac{0.05\times 0.95}{N}} \approx 0.0022$
- Indeed, for the values of $n = 10; 20; 30; 50$, the estimate of α is much lower than the nominal value of 5%; this shows that the resulting test is quite conservative.
- **Conclusion: the approximation to asymptotic results is not suitable for moderate values of n.**

Example (cont): Asymmetry test for the normality

We know that, under normality of the model, the exact distribution of the asymmetry coefficient $\sqrt{b_1} \sim N(0, \frac{6(n-2)}{(n+1)(n+3)})$

In this case we just have to substitute the critical points by the quantiles of order $(1 - \frac{\alpha}{2})$ of $N(0, \frac{6(n-2)}{(n+1)(n+3)})$

```
cv <- qnorm(.975, 0, sqrt(6*(n-2)/((n+1)*(n+3))))

> round(cv,4)
[1] 1.1355 0.9268 0.7943 0.6398 0.4660 0.2134
```

After repeating the simulation for this vector cv of new critical points we get the results:

n	10	20	30	50	100	500
probERRO1	0.0536	0.0516	0.0565	0.0531	0.0505	0.0505

Conclusion: these estimates are much closer of significance level $\alpha = 0.05$

Exercise: critical points simulated for $\sqrt{b_1}$

Compare through a histogram of simulated PRN's from the asymmetry coefficient$\sqrt{b_1}$, supposing normality, overlapping the asymptotic distribution pdf of the same coefficient, considering the exact or the approximate variance correction.

Simulate the critical points for the normality test formulated by

$$H_0 : \sqrt{\beta_1} = 0 \quad vs \quad H_1 : \sqrt{\beta_1} \neq 0$$

and compare with critical points approximated to the normal, with(without the variance correction.

For $n = 500, \alpha = 5\%, N = 100000$, next there's an algorithm and the required plots:

3. Applications in Statistical Inference: statistical tests

```
# Pontos críticos simulados de sqrt_b1, n=500

n=500
N=100000
sqrt_b1<-numeric()
for (j in 1:N) {
x <- rnorm(n)
sqrt_b1[j]<-sk(x)
}

hist(sqrt_b1, prob = TRUE,main = NULL,
    xlim=c(-0.4,0.4),ylim=c(0,4.8))
text(0.0,4.8, "Histograma para o coeficiente
     de assimetria, n=500")
text(0.0,4.5, "N=100000")

t<-seq(-1.5,1.5,by=0.001)
lines(t,dnorm(t,0,sqrt(6*(n-2)/((n+1)*(n+3)))),
    lty=1,type="l",xlim=c(-1,1),
        ylim=c(0,5),col="red")
text(0.2,3.7,"fdp N(0,6(n-2)/((n+1)(n+3)))",
    col="red")

lines(t,dnorm(t,0, sqrt(6/n)),lty=1,type="l",
    xlim=c(-1,1),ylim=c(0,5),col="blue")
text(0.15,3., "fdp N(0,6/n)",col="blue")
```

```
SkCoef<-sort(sqrt_b1)

q025<- floor(N*.025)+1
q975<- floor(N*.975)+1

# quantil_.025  do Coef Assimetria sqrt_b1 simulado
q025<- SkCoef[q025]
# quantil_.975  do Coef Assimetria sqrt_b1 simulado
q975<- SkCoef[q975]

# quantil_.025 da N(0,6(n-2)/((n+1)(n+3)))
  z1025 <- qnorm(.025, 0, sqrt(6*(n-2)/((n+1)*(n+3))))
# quantil_.975 da N(0,6(n-2)/((n+1)(n+3)))
  z1975 <-  qnorm(.975, 0, sqrt(6*(n-2)/((n+1)*(n+3))))

# quantil_.025 da N(0,6/n)
  z2025 <- qnorm(.025, 0, sqrt(6/n))
# quantil_.025 da N(0,6/n)
  z2975 <-  qnorm(.975, 0, sqrt(6/n))

 cat("quantis assintóticos","\n",
 "   z2_025=",  z2025, "   z2_975=", z2975,"\n",
 "quantis assintóticos com variância exacta","\n",
 "   z1_025=",  z1025, "   z1_975=", z1975,"\n",
 "quantis exactos de sqrt_b1 simulados","\n",
  "   q025=",  q025, "   q975=", q975,"\n"     )
```

asymptotic quantiles

$z2_{0.025} = -0.2147033 \quad z2_{0.975} = 0.2147033$

asymptotic quantiles with exact variance

$z1_{0.025} = -0.2134202 \quad z1_{0.975} = 0.2134202$

exact quantiles of simulated values of $\sqrt{b_1}$

$q_{0.025} = -0.2138422 \quad q_{0.975} = 0.2142118$

